

Figure 1: Streams

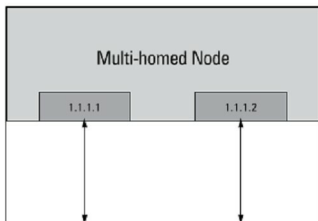


Figure 2: An example of a multi-homed node

streams. SCTP streams are unidirectional—see Figure 1.

This reduces the delay caused by 'head-of-line' blocking experienced in TCP, since there is an independence of delivery across 'streams' within an association. Even programmatically, one finds it useful to separate logically different streams of data over the same socket, and SCTP provides this facility. For those of you who exclaimed, "This brings the same advantage to data transmission that threads brought to process management!"—you are right...more or less.

Multi-homing

A network element can be called multi-homed if it can be addressed by, and thus 'owns' multiple IP addresses—see Figure 2.

When a peer is multi-homed, one of the IP addresses is selected as the 'primary destination address'. All data will be destined to this IP, by default. When a failure of a primary address is detected, the sender selects an alternate primary address until the link is restored or the user initiates a change in the primary address.

During the process of association establishment, an SCTP endpoint can (optionally) provide a list of transport addresses (i.e., multiple IP addresses in combination with an SCTP port—thus making it multi-homed), through which that endpoint can be reached, and from which



Figure 3: Example of multi-homing during network failure

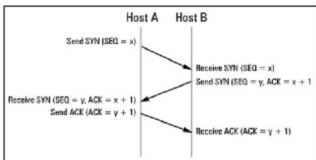


Figure 4: A three-way TCP handshake

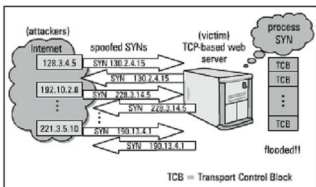


Figure 5: A SYN attack

it will originate SCTP packets. The association covers transfers over all of the possible source/destination combinations that may be generated from each endpoint's lists.

This provides a better chance of the session surviving during network failures. In a conventional single-homed session, the failure of local network access can isolate the peer. Refer to Figure 3.

SCTP provides this feature, enabling sessions to be more network-fault tolerant.

Associations—more than connections

SCTP is connection-oriented in nature. An 'SCTP association' is a broader concept than a TCP connection, since it incorporates the multi-streaming facility and multi-homing capability.

The SCTP handshake mechanism

We are aware of the TCP 3-way handshake (see Figure 4). We are also aware of how this leads to SYN attacks, wherein a server is brought to its knees, and refuses to accept any more connections (a victim of DOS attacks). Figure 5 briefly illustrates how it happens.